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A novel approach to sensitivity evaluation of laser-induced breakdown spectroscopy for rare earth elements determination

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We report the potential of Laser-Induced Breakdown Spectroscopy (LIBS) for the determination of lanthanum and yttrium in soils and rocks. Since the main problem to quantify rare earth elements by LIBS is their rich spectra and the consequent frequent spectral interferences with the matrix elements, we demonstrated how thermodynamic modeling of the spectra can assist spectroscopists in the estimation of LIBS sensitivity. The theoretical LOD for La was close to the one retrieved from experimental data (6 ppm), while the theoretical LOD for Y was one order of magnitude higher than the experimental LOD (6 ppm *vs.* 0.4 ppm). The possible reasons for such a discrepancy are discussed.

Laser-Induced Breakdown Spectroscopy (LIBS) uses laser ablation for generation of the plasma, which emission serves for the analysis of a sample. In recent years, LIBS has proved to be efficient for rapid real time *in situ* analysis of environmental samples, including a variety of geological samples such as soils, rocks or even gemstones.[1–4] The specific feature of LIBS analysis in geochemical studies consists in obtaining the fingerprints of geological objects. Different classes of minerals can be discriminated[5] on the basis of the emission lines of major components, but it is more difficult to identify correctly a particular mineral within the same type of sample with a fixed major element composition and a high degree of trace element variability. Moreover, geological samples contain ten to twelve major elements including elements with rich emission spectra (iron, titanium, *etc.*) providing strong spectral interferences. Thus, accurate trace element analysis is quite a challenge.

Rare earth elements (REEs) are among the main markers in geochemistry. REE are members of group 3 of the periodic table and all of them have similar physical and chemical properties, but these elements may be partially separated by some petrological and mineralogical processes.[6] To our best knowledge there are no works on quantitative REE determination by LIBS in geological solid samples due to the significant spectral interferences and relatively low content of REEs in the main geochemical materials (soils, sediments and rocks). And there are only few works on REE quantitative LIBS analysis in aqueous solutions[7,8] or in synthetic simple matrices.[9–12] Thus, an evaluation of the achievable sensitivity of LIBS determination of the REEs in solid natural samples is highly beneficial.

We present an approach to estimate the limit of detection (LOD) with the use of thermodynamic modeling of LIBS spectra. This method is based on generating synthetic spectra calculated under local thermodynamic equilibrium (LTE) conditions and sensitive to the Doppler and Stark broadenings of emission lines in laser plasma. Spectral slit width as well as diffraction function of a spectrometer, and detector instrumental function (combination of an image intensifier resolution and spectral CCD pixel width) are fixed to an instrument specification. Assuming that the detection capability is restricted mainly by two factors, namely signal-to-noise ratio (SNR) and spectral interferences, we estimated LODs with the use of experimental noise and synthetic spectra calculated for various ratios analyzed for the interfering species. This also enables estimation of the accuracy of LIBS analysis with the use of small number of certified samples. We examined light REEs (yttrium, and lanthanum) in soils and rocks by LIBS to validate our approach.

In our previous work,[13] we used the simple model of optically thin homogeneous plasma under LTE conditions to calculate emission spectra. Although a condition providing optically thin plasma is preferable for LIBS analysis,[14] it is difficult to realize these for species of the main components especially for intense lines.[15] Overlapped wings of such broadened self-absorbed lines significantly influence the shape of a spectrum, making a pseudo-background and masking weak lines of trace components. In the current study the possible self-absorption of the emission lines was accounted for to provide more realistic synthetic spectra. We used the so-called transport equation and assumption of a homogenous plasma rod for the calculation. In this case, light intensity corresponds to the energy emitted per

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Table 1 CRM sample composition

Sample	Type	La (ppm)	Y (ppm)
SChT-1 (2507-83)	Soil	36 ± 5	31 ± 6
BIL-1 (7126-94)	Sediment	45 ± 6	30 ± 4
NIST 2710a	Soil	30.6 ± 1.2	n/a
NIST 2711a	Soil	38 ± 1	n/a
SSV-1 (6104-91)	Rock	108 ± 17	25 ± 5

Table 2 Limits of detection of La and Y in soils and ores

	LOD (theoretical), ppm		LOD (experimental), ppm	
Element	Peak	Integral	Peak	Integral
La	6.5	4.1	8.0	4.8
Y	6.0	6.0	0.6/3.0[a]	0.4/2.5[a]

[a] The noise level was calculated on symmetrical wing of Si I 410.29 nm line.

unit time, surface and wavelength by a rod of certain length. The equation for the calculation of the absorption coefficient as a function of wavelength and plasma rod length, which takes into account both the absorption by lower level atoms and the stimulated emission of the upper level atoms under the effect of the radiation, was taken from ref. 16. Thus, in the current study we used a static model of homogeneous plasma under LTE conditions, accounting for instrumental function, Stark and Doppler broadening as well as self-absorption.

For the experimental study we used US and Russian certified reference materials listed in Table 1 and their full composition is available elsewhere.[17] The experimental setup was the same as that described earlier by us.[18] The second harmonic Q-switched Nd:YAG laser was focused on pelleted samples, providing a power density ~3–6 GW cm^{-2} on the surface. Spectra were recorded by a Czerny–Turner spectrometer equipped with an ICCD camera. The best signal-to-noise ratio was achieved at a time delay 1.5 μs and gate 1.5 μs.

Initially we assessed the La determination in soils and rocks as a relatively simple case due to the presence of the isolated ionic line of La II at 433.37 nm in the LIBS spectrum of an SSV-1 sample (Fig. 1a). We prepared a series of synthetic spectra of the plasma (430.60–435.60 nm) with the same elemental composition as the SSV-1 sample. The plasma parameters were varied over the intervals $0.5 < T$, eV < 1.0 and $15 < \lg(N_e$, cm$^{-3}) < 18$. The best matched pair of the model and experimental spectra (Fig. 1a) was found by the maximal value of Pearson's correlation coefficients between them as described in ref. 13. We normalized both spectra by La line intensity to compare the synthetic and experimental data. The plasma parameters for the best correlated synthetic spectrum were $T = 0.72$ eV and $\lg(N_e$, cm$^{-3}) = 16.5$, and they are in a good agreement with experimental data on soil ablation.[19] We prepared a common linear calibration curve for La ($r^2 = 0.968$) to estimate the sensitivity in terms of LOD with the use of 3σ-criteria. The noise level was estimated in the spectral range without emission lines near the La II line at 433.37 nm, which is indicated in Fig. 1a. We used both the integral and peak intensity of the La line to achieve the experimental values of the LODs (see Table 2). We used the synthetic spectra (Fig. 1b) and experimental values of the noise to estimate the theoretical LOD values, also with the use of 3σ-criteria. As expected, the experimental LODs and theoretical ones are close to each other. This means that the background emission is negligible and the La II line is well isolated without spectral interferences. Slightly lower values of LOD in the case of the synthetic spectra can be related to a more correct presentation of the line shape at low intensities.

A more complicated task was Y determination in soils and ores. Following the NIST database,[20] the Y I 408.37 nm line has the highest relative intensity, but it turned out that it was not sensitive to Y variations in soils and rocks. We have prepared synthetic spectra of soil sample SChT-1 to explain such a behavior (Fig. 2a). We obtained another set of synthetic spectra within the spectral range 408.0–412.7 nm. The plasma parameters ($T = 0.65$ eV and $\lg(N_e$, cm$^{-3}) = 16.6$) of synthetic spectra provided the highest correlation with experimental one. To recognize the possible spectral interferences, the line intensities obtained before the convolution procedure are added to Fig. 2 as the bars. The Y I line is significantly weaker than the line of Mn I at 408.36 nm (Mn content in the sample is 612 ppm), which completely suppressed the Y signal.

Thereafter we studied another strong line of Y I at 410.24 nm (Fig. 2b). It can be resolved in spite of the possible overlap with

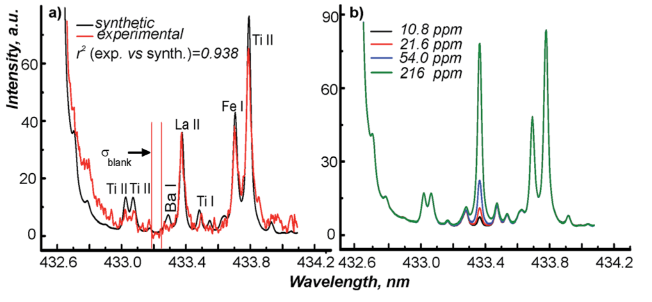

Fig. 1 Emission spectra near La II 433.37 nm: (a) comparison between experimental spectrum for a SSV-1 sample and the best fitted synthetic spectrum; (b) a set of synthetic spectra with variations in La content.

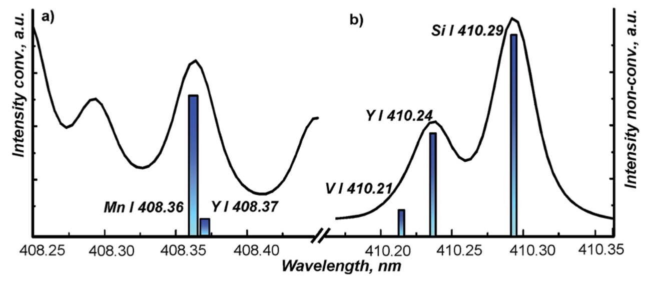

Fig. 2 Synthetic spectra for SChT-1 soil near to Y I 408.37 nm (a) and Y I 410.24 nm (b). The position and height of the bars indicate the wavelength and intensity of the neighboring lines, respectively.

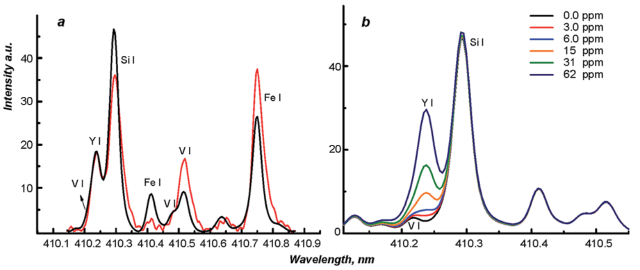

Fig. 3 Emission spectra near Y I at 410.24 nm: (a) comparison of the experimental spectrum obtained for SChT-1 (red curve) and the synthetic spectrum (black curve); (b) a set of synthetic spectra near the line of Y I at 410.24 nm with the Y content variation.

neighboring lines. The experimental spectrum and the synthetic one obtained under the above mentioned plasma parameters are compared in Fig. 3a. It should be noted that the synthetic spectrum revealed significant spectral interferences of the Si I line at 410.29 nm (silicon is the main component of soils) and the V I line at 410.21 nm (V content in the sample is 72 ppm) at low concentrations of yttrium (Fig. 3b). Similarly to La, we calculated experimental LODs for Y determination (Table 2). We calculated the intensity of the Y line after simple subtraction of the baseline. This led to the upper estimation of LOD, lowering the Y signal. The use of an upper limit seems appropriate, since we used linear extrapolations to the low concentrations. Besides the spectral range without emission lines, the symmetrical wing of the Si I line near 410.34 nm was used for the noise estimation to account for the fluctuations of both background and interfering lines. As one would expect, the LOD values were much higher (by 5–6 times) in the latter case. The theoretical LODs of Y determination in soils and rocks were evaluated as close to 6 ppm, when the Y signal can be distinguished even without noise consideration. We estimated equal values of LOD for the peak and integral intensities, since it is impossible to single out the Y line and, as a consequence, calculate its integral intensity even with the deconvolution procedure below 6 ppm of Y. The LODs calculated on the experimental calibration curve are significantly lower than theoretical LODs. A clear discrepancy between two sets of LODs for yttrium seems to be a result of an extrapolation to low concentrations that did not allow for a correct consideration of spectral interferences.

Conclusions

This work demonstrates that the thermodynamic modeling of the emission spectra of a laser plasma can be fruitful to plan LIBS analytical studies prior to experimental measurement to evaluate the possible spectral interferences both from weak lines of major components and strong lines of minor and trace elements. The comparison of synthetic and experimental data provides a more correct evaluation of the LIBS sensitivity and of the LOD calculation, because the spectral interferences from the weak lines should be accounted for in the extrapolation to a low content of the analyzed element. This possibility is very important for the analysis of natural solid samples, for which the lack of certified materials is still a serious problem.[21] Finally, our study demonstrates that LIBS can be used for the determination of light rare earth elements, like Y and La, under geological exploration since the achieved LODs are sufficiently lower than their Earth crustal abundances: 21 ppm for Y and 31 ppm for La.[22]

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References

1 R. S. Harmon, R. E. Russo and R. R. Hark, *Spectrochim. Acta, Part B*, 2013, **87**, 11.
2 G. S. Senesi, *Earth-Sci. Rev.*, 2014, **139**, 231.
3 S. Qiao, Y. Ding, D. Tian, L. Yao and G. Yang, *Appl. Spectrosc. Rev.*, 2015, **50**, 1.
4 N. B. Zorov, A. M. Popov, S. M. Zaytsev and T. A. Labutin, *Russ. Chem. Rev.*, 2015, **84**, 1021.
5 R. S. Harmon, J. Remus, N. J. McMillan, C. McManus, L. Collins, J. L. Gottfried Jr, F. C. DeLucia and A. W. Miziolek, *Appl. Geochem.*, 2009, **24**, 1125.
6 P. Henderson, *Rare Earth Element Geochemistry*, Elsevier Science, 1983.
7 P. Fichet, M. Tabarant, B. Salle and C. Gautier, *Anal. Bioanal. Chem.*, 2006, **385**, 338.
8 D. Alamelu, A. Sarkar and S. K. Aggarwal, *Talanta*, 2008, **77**, 256.
9 T. Ishizuka, *Anal. Chem.*, 1973, **45**, 538.
10 L. C. Jensen, S. C. Langford, J. T. Dickinson and R. S. Addleman, *Spectrochim. Acta, Part B*, 1995, **50**, 1501.
11 A. Weisberg, R. E. Lakis, M. F. Simpson, L. Horowitz and J. Craparo, *Appl. Spectrosc.*, 2014, **68**, 937.
12 M. Martin, R. C. Martin, S. Allman, D. Brice, A. Wymore and N. Andre, *Spectrochim. Acta, Part B*, 2015, **114**, 65.
13 T. A. Labutin, S. M. Zaytsev and A. M. Popov, *Anal. Chem.*, 2013, **85**, 1985.
14 D. A. Cremers and L. J. Radziemski, *Handbook of Laser-Induced Breakdown Spectroscopy*, John Wiley & Sons, Ltd., 2006.
15 C. Aragón, F. Peñalba and J. A. Aguilera, *Spectrochim. Acta, Part B*, 2005, **60**, 879.
16 F. Bredice, F. O. Borges, H. Sobral, M. Villagran-Muniz, H. O. Di Rocco, G. Cristoforetti, S. Legnaioli, V. Palleschi, L. Pardini, A. Salvetti and E. Tognoni, *Spectrochim. Acta, Part B*, 2006, **61**, 1294.
17 K. P. Jochum, U. Nohl, K. Herwig, E. Lammel, B. Stoll and A. W. Hofmann, *Geostand. Geoanal. Res.*, 2005, **29**, 333.
18 S. M. Zaytsev, A. M. Popov, N. B. Zorov and T. A. Labutin, *J. Instrum.*, 2014, **9**, P06010.
19 I. Rauschenbach, V. Lazic, S. G. Pavlov, H.-W. Hübers and E. K. Jessberger, *Spectrochim. Acta, Part B*, 2008, **63**, 1205.

20 A. Kramida, Y. Ralchenko, J. Reader and NIST ASD Team, *NIST Atomic Spectra Database (ver. 5.3)*, 2015, http://physics.nist.gov/asd.
21 O. T. Butler, W. R. L. Cairns, J. M. Cook and C. M. Davidson, *J. Anal. At. Spectrom.*, 2016, **31**, 35.
22 R. L. Rudnick and S. Gao, Composition of the Continental Crust, in *Treatise on Geochemistry, The Crust*, ed. R. L. Rudnick, Elsevier, 2nd edn, 2014, ch. 4.1, vol. 4.

